McKell Woodland

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Education

Ph.D. in Computer Science, Rice University, Houston, TX, USA 2024

Thesis Title: Predicting Liver Segmentation Model Failure with Feature-Based Out-of-Distribution Detection and Generative Adversarial Networks

Thesis Advisor: Kristy K. Brock

B.S. in Applied Mathematics, Brigham Young University, Provo, UT, USA 2018

Publications

Peer-Reviewed Articles

J. Chun, **M. Woodland**, A. Castelo, C. O'Connor, M. Altaie, A. Gupta, S. Ding, E. Koay, & K.K. Brock (2026). Context-Informed Personalized Segmentation from Radiology Reports: A Fully Automated Framework for Long-term Liver Tumor Follow-Up. *Med. Phys.* In press.

J. Chun, A. Castelo, **M. Woodland**, C.S. O'Connor, M. Altaie, M. Eltaher, A.C. Gupta, B. Daoud, S. Ding, J. Bennett, A. Maitra, M.A. Firpo, K. Kirkwood, E.J. Koay, & K.K. Brock (2025). Uncertainty-Guided Test-Time Optimization for Personalizing Segmentation Models in Longitudinal Medical Imaging. *Med. Phys.*, 53(1), e70206. doi.org/10.1002.mp.70206

M.E. Woodland, M. Altaie, C.S. O'Connor, A.H. Castelo, O.C. Lebimoyo, A.C. Gupta, J.P. Yung, P.E. Kinahan, C.D. Fuller, E.J. Koay, B.C. Odisio, A.B. Patel, & K.K. Brock (2025). Generative Modeling for Interpretable Anomaly Detection in Medical Imaging: Applications in Failure Detection and Data Curation. J. *Bioeng.*, 12(10), 1106. doi.org/10.3390/bioengineering12101106

M. Woodland, N. Patel, A. Castelo, M. Al Taie, M. Eltaher, J.P. Yung, T.J. Netherton, T.L. Calderone, J.I. Sanchez, D.W. Cleere, A. Elsaiey, N. Gupta, D. Victor, L. Beretta, A.B. Patel, & K.K. Brock (2024). Dimensionality Reduction and Nearest Neighbors for Improving Out-of-Distribution Detection in Medical Image Segmentation. *MELBA*, 2(UNSURE2023 special issue), 2006-2052. doi.org/10.59275/j.melba.2024-g93a

H. Baroudi, K.K. Brock, W. Cao, X. Chen, C. Chung, L.E. Court, M.D. El Basha, M. Farhat, S. Gay, M.P. Gronberg, A.C. Gupta, S. Hernandez, K. Huang, D.A. Jaffray, R. Lim, B. Marquez, K. Nealon, T.J. Netherton, C.M. Nguyen, B. Reber, D.J. Rhee, R.M. Salazar, M.D. Shanker, C. Sjogreen, **M. Woodland**, J. Yang, C. Yu, & Y. Zhao. (2023). Automated Contouring and Planning in Radiation Therapy: What Is 'Clinically Acceptable'? *Diagnostics*, 13, 667. doi.org/10.3390/diagnostics13040667

Sen, P. Troncoso, A. Venkatesan, M.D. Pagel, J.A. Nijkamp, Y. He, A. Lesage, **M. Woodland**, & K.K. Brock (2021). Correlation of In-Vivo Imaging with Histopathology: A Review. *Eur. J. Radiol.*, 144, 109964. doi.org/10.1016/j.ejrad.2021.109964

M. Woodland, M. Hsiung, E. Matheson, A. Safsten, J. Greenwood, D.M. Halverson, & L.L. Howell (2021). Analysis of the Rigid Motion of a Developable Conical Mechanism. *Mech. Robot*, 13, 031106. doi.org/10.1115/1.4050294

R. Kimmons, E. Hunsaker, E.J. Jones, & **M. Stauffer** (2019). The nationwide landscape of K-12 school websites in the United States: Systems, services, intended audiences, and adoption patterns. *IRRODL*, 20(3). doi.org/10.19173/irrodl.v20i4.3794

R. Kimmons, K. McGuire, **M. Stauffer**, E.J. Jones, M. Gregson, & M. Austin (2017). Religious identity, expression, and civility in social media: Results of data mining Latter-day Saint Twitter accounts. *JSSR*, 56(3), 180-210. doi.org/10.1111/jssr.12358

Conference Proceedings

X. Zhang, C. O'Connor, M. Woodland, A. Castelo, I. Paolucci, J. Albuquerque Marques Silva, O. Prabhugaonkar, A. Patel, & K.K. Brock. Limited-FOV Liver Deformable Registration via Transformer-Based Point-Cloud Completion. *CAPTION 2026* (MICCAI workshop). In press.

M. Nielsen, A. Castelo, M. Altaie, A. Anthony, N. Siddiqi, K.K. Brock, & **M. Woodland**. Label-Free Threshold Selection for Out-of-Distribution Detection in Liver CT Segmentation. *UNSURE 2026* (MICCAI workshop). In Press.

J. Bennett*, **M. Woodland***, A. Castelo, M. Altaie, A. Anthony, N. Siddiqi, J.P. Long, & K.K. Brock. AI Failure Detection in Clinical CT Liver Segmentation: A Large-Scale Evaluation. *MICCAI 2026*. In press. * equal contribution

M. Woodland, A. Castelo, M. Al Taie, J. Albuquerque Marques Silva, M. Eltaher, F. Mohn, S. Kundu, J.P. Yung, A.B. Patel, & K.K. Brock (2024). Feature Extraction for Generative Medical Imaging Evaluation: New Evidence Against an Evolving Trend. *MICCAI 2024*. LNCS, 15012. Springer, Cham. doi.org/10.1007/978-3-031-72390-2_9

M. Woodland, N. Patel, M. Al Taie, J.P. Yung, T.J. Netherton, A.B. Patel, & K.K. Brock (2023). Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. *UNSURE 2023* (MICCAI workshop). LNCS, 14291. Springer, Cham. doi.org/10.1007/978-3-031-44336-7_15

M. Woodland, J. Wood, B.M. Anderson, S. Kundu, E. Lin, E. Koay, B. Odisio, C. Chung, H.C. Kang, A.M. Venkatesan, S. Yedururi, B. De, Y.-M. Lin, A.B. Patel, & K.K. Brock (2022). Evaluating the Performance of StyleGAN2-ADA on Medical Images. *SASHIMI 2022* (MICCAI workshop). LNCS, vol 13570. Springer, Cham. doi.org/10.1007/978-3-031-16980-9_14

M. Woodland, M. Hsiung, E.L. Matheson, C.A. Safsten, J. Greenwood, D.M. Halverson, & L.L. Howell (2020). Analysis of the Rigid Motion of a Developable Conical Mechanism. *IDETC-CIE 2020*. doi.org/10.1115/DETC2020-22643

A. Henrichsen, B. Jafek, J. O'Bryant, & **M. Stauffer** (2018). Brigham Young University Speeches Popularity Predictor. *NCUR 2018*. libjournals.unca.edu/ncur/2018-2/

Under Review

X. Zhang, C.S. O'Connor, A.H. Castelo, **M.E. Woodland**, B. Daoud, I. Paolucci, J. Albuquerque Marques Silva, N.S. Siddiqi, B.C. Odisio, A.B. Patel, & K.K. Brock. BioDeformUNet: A Deep Learning Model for Biomechanically Informed Liver Image Registration.

A. Castelo, C. O'Connor, A.C. Gupta, B.M. Anderson, **M. Woodland**, M. Altaie, E.J. Koay, B.C. Odisio, T.T. Tang, & K.K. Brock. Quality versus quantity of training datasets for artificial intelligence-based whole liver segmentation.

Selected Abstracts

M. Woodland, C. O'Connor, J. Wood, A.B. Patel, & K.K. Brock (2023). Interpretable Out-of-Distribution Detection with Generative Adversarial Networks. *Med. Phys.*, 50(6), e154-155. doi.org/10.1002/mp.16525

M. Woodland, J. Wood, C. O'Connor, A.B. Patel, & K.K. Brock (2022). StyleGAN2-based Out-of-Distribution Detection for Medical Imaging. In: *Med-NeurIPS 2022*. sites.google.com/view/med-neurips2022/abstracts

B. Reber, L.V. van Dijk, B.M. Anderson, A.S. Mohamed, B. Rigaud, Y. He, **M. Woodland,** C.D. Fuller, S.Y. Lai, and K.K. Brock (2022) Comparison of Machine Learning and Deep Learning Methods for the Prediction of Osteoradionecrosis Resulting from Head and Neck Cancer Radiation Therapy. *Int. J. Radiat. Oncol. Biol. Phys.*, 114(3), e124. doi.org/10.1016/j.ijrobp.2022.07.946

M. Woodland, J. Wood, B.M. Anderson, S. Kundu, E. Lin, E. Koay, B. Odisio, C. Chung, H.C. Kang, A. Venkatesan, A., S. Yedururi, B. De, Y. Lin, A.B. Patel, & K.K. Brock (2022). Comparing Transfer Learning, Data Augmentation, and Data Expansion in the Improvement of Medical Image Generation. *Med. Phys.*, 49(6), 134. doi.org/10.1002/mp.15769

M. Woodland, A.B. Patel, B.M. Anderson, E. Lin, E. Koay, B. Odisio, & K.K. Brock (2021). GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *Med. Phys.*, 48(6). doi.org/10.1002/mp.15041

Research Experience

Data Scientist; Graduate Research Assistant

2020 – Present

The University of Texas MD Anderson Cancer Center, Houston, TX, USA — Supervisor: Dr. Kristy Brock

Focused on detecting failures in AI-based segmentation models for clinical use. Proposed an accurate feature-based out-of-distribution detection methodology with sub-second performance that is broadly applicable post-model development. Investigated generative modeling for interpretable failure detection, evaluated the application of StyleGAN2 to medical imaging, and contributed to best practices for evaluating generative models in medical imaging.

Student Researcher
Fall 2018

Rice University, Houston, TX, USA — Supervisor: Drs. Craig Rusin & Parag Jain

Developed a real-time monitoring algorithm using CNN-based morphologic parametrization of EKG signals to detect postoperative junctional ectopic tachycardia with 87% sensitivity and specificity.

Graduate Research Assistant
2018 – 2019

Brigham Young University, Provo, UT, USA — Supervisor: Dr. David Wingate

Improved reinforcement learning efficiency by embedding game states, reducing sample complexity without compromising accuracy.

Research Assistant
Summer 2017

Brigham Young University, Provo, UT, USA — Supervisor: Dr. Tyler Jarvis

Explored the use of Markov chains to optimize facility location problems, introducing a novel construction applied to determining the ideal site for a new religious center.

Research Assistant
Summer 2016

Brigham Young University, Provo, UT, USA — Supervisor: Dr. Denise Halverson

Analytically constructed a developable mechanism on a conical surface with rigid motion, demonstrating inherent bifurcation points leading to multiple motion trajectories.

Research Assistant
2016

Brigham Young University, Provo, UT, USA — Supervisor: Dr. Royce Kimmons

Conducted a nationwide analysis of K–12 school websites to examine adoption patterns. Analyzed tweets from projected Latter-Day Saint accounts to examine religious identity and civility.

Awards

Outstanding Reviewer – Honorable Mention, *MICCAI 2025*, highly-rated reviews — 2025

Outstanding Reviewer – Honorable Mention, *MICCAI 2024*, highly-rated reviews — 2024

Best Spotlight Paper, *UNSURE 2023 (MICCAI workshop)*, top spotlight presentation — 2023

Graduate Student Travel Grant, *Rice Engineering Alumni*, travel award to UNSURE 2023 — 2023

Early Career Investigator – Significance, *PBDW 2021*, early career research with the most significance — 2021

Worldwide Innovations in Medical Physics Scholarship, *WIMP 2021*, early career competition winner — 2021

Showcase Winner, *Rice Data2Knowledge Showcase*, top research project — 2019

Session Winner, *BYU Student Research Conference*, top research presentation — 2019

Exceptional Leader, *BYU Y-serve*, top leadership award for founding Kids Who Code — 2019

Talks & Presentations

Invited Talks

Detection and Impact of Distribution Shifts on Radiology AI Models. *RSNA 2025*.

Predicting Segmentation Model Failure with Out-of-Distribution Detection and Generative Adversarial Networks. *Tumor Measurement Initiative Symposium 2025*.

Predicting Liver Segmentation Model Failure with Feature-based Out-of-Distribution Detection and Generative Adversarial Networks. *Computer Science Colloquia at BYU, 2024.*

Oral Presentations

Label-Free Threshold Selection for Out-of-Distribution Detection in Liver CT Segmentation. *UNSURE 2026* (MICCAI workshop)

Predicting Liver Segmentation Model Failure with Feature-Based Out-of-Distribution Detection and Generative Adversarial Networks. *MICCAI Student Board PhD Thesis Madness 2024*.

Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. *UNSURE 2023* (MICCAI workshop).

Interpretable Out-of-Distribution Detection with Generative Adversarial Networks. *AAPM 2023*.

Comparing Transfer Learning, Data Augmentation, and Data Expansion in the Improvement of Medical Image Generation. *AAPM 2022*.

Houston, Our AI Models Have a Problem! *SWAAPM 2022*.

GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *AAPM 2021*.

Understanding LDS Twitter use. *Instructional Design and Learning Community Conference 2016*.

Selected Poster Presentations

Feature Extraction for Generative Medical Imaging Evaluation: New Evidence Against an Evolving Trend. *MICCAI 2024*.

Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. *UNSURE 2023 (MICCAI Workshop)*.

Interpretable OOD Detection using GANs: An Application to the MIDRC Data Repository. *PBDW 2023.*

StyleGAN2-based Out-of-Distribution Detection for Medical Imaging. *Med-NeurIPS 2022 (NeurIPS workshop).*

Evaluating the Performance of StyleGAN2-ADA on Medical Images. *SASHIMI 2022 (MICCAI workshop).*

StyleGAN2-based Out-of-Distribution Detection in Computed Tomography. *PBDW 2022.*

Improving the Generation of Synthetic Medical Images using Data Augmentation and Transfer Learning. *SWAAPM 2022.*

Detecting Out-of-Distribution Images for Active Learning using a Generative Adversarial Network (StyleGAN2). *PBDW 2021.*

GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *WIMP 2021.*

A Novel Algorithm for Early Detection of Junctional Ectopic Tachycardia in Patients with Congenital Heart Disease. *WFPICCS 2020.*

Professional Experience

Microsoft, Redmond, WA, USA (remote) — Summer 2020

Data Science Intern

Developed a similarity score to analyze the overlap between suspicious behavior detectors for Microsoft Threat Experts.

Lawrence Livermore National Laboratory, Livermore, CA, USA — Summer 2019

Data Science Research & Development Intern

Engineered an event detection model for video streams that was optimized for identifying obscured and moving objects.

3M Health Information Systems, Murray, UT, USA — Summer 2018

Data Science Intern

Designed deep learning frameworks to predict over 50,000 diagnoses and DRG codes from 1.7 million clinical documents.

National Security Agency, Laurel, MD, USA — Summer 2017

Director's Summer Program Intern

Investigated an inertial navigation problem with noisy data, authored an internal technical paper, and delivered a briefing to the Deputy Director of the NSA.

Teaching Experience

Lead Instructor; Developer — 2019 - 2024

The Coding School, Remote

Instructed 300+ HBCU and community college professors on how to introduce AI into their classrooms. Authored the deep learning and SQL curriculum and edited the machine learning curriculum.

Online Adjunct Instructor Summer 2022

Computer Science, Brigham Young University – Idaho, Remote

Piloted the online version of the course *"Machine Learning and Data Mining"*.

Teaching Assistant 2020-2021

Computer Science, Rice University, Houston, TX, USA

Courses TA'd: *Statistics for Computing and Data Science*, *Introduction to Computer Security*, *Programming for Data Science*

Instructional Coach 2021-2022

Computer Science, Rice University, Houston, TX, USA

Helped students prepare to give research presentations in the department's Graduate Research Seminar.

Curriculum Developer 2017

Applied Mathematics, Brigham Young University, Provo, UT, USA

Wrote 7 Python labs (*Iterative Solvers*, *Profiling*, *SymPy*, *SQL 1*, *SQL 2*, *One-Dimensional Optimization*, *Gradient Descent*) and edited over 15 other labs.

Academic Services

Moderation

2nd Annual Research Retreat in Oncological Data and Computational Sciences, 2022.

Internal Q&A session for interns with the CFO of Microsoft's AI Cognitive Services, 2020.

Panel Participation

Best Practices for AI Continuous Model Evaluation. *RSNA 2025*.

Panel Discussion on the Trips and Tips of MICCAI Review, Rebuttal, Challenge, and Workshop. *MICCAI Student Board Webinar Series*, 2024.

"And You're Out!" – Setting Minimum Expectations for Big Data/AI Abstracts. *PBDW 2023.*

Reviewing

MICCAI, 2024-2026

International Journal of Computer Assisted Radiology and Surgery, 2025-2026

Nature Communications, 2025

Sensors, 2025

Applied Sciences, 2025

Communications Medicine, 2024

UNSURE (MICCAI workshop), 2024

Deep Generative Models for Health (NeurIPS workshop), 2023

Outreach

Young Professional Advisory Board Member, *The Coding School*, designed governance structure 2024 – Present

Speaker, *Brigham Young University*, invited to speak on medical AI research in a careers class 2025

Instructor, *The Coding School*, taught Python/AI to 400+ primary school students 2020-2024

Mentor, *Rice Datathon*, mentored undergraduate participants 2022

Teaching Assistant, *Rice University*, tutored online master's students who were struggling with curriculum 2022

Research Mentor, *Summer STEM Institute*, advised 2 high school students on medical AI research 2021

Speaker, *Summer STEM Institute*, delivered talk to 300+ students on medical AI research 2021

Panelist, *Rice CS/IO Club*, spoke on graduate student panel 2021

Executive Director, *BYU Y-Serve: Kids Who Code*, founded organization to teach coding to local schools 2018-2019

Ambassador, *Utah STEM Action Center*, promoted STEM funding during 10+ events and 100+ calls 2018-2019

Facilitator, *Girls Who Code*, taught 20+ girls at an elementary school Scratch 2019

Panelist, *Girls Who Code*, answered questions about coding at event with 200+ middle school girls 2018

Public Outreach Officer*, BYU Women in Computer Science,* planned hackathon; ran social media 2018